Hanzhi Zhang (she/her)

Ph.D. Candidate in Computer Science | University of North Texas, Denton, TX, US

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RESEARCH INTERESTS

Efficient and Sustainable AI Systems: Data–model–hardware co-design, hardware-aware LLM inference, sparse and mixed-precision computation, post-training quantization, GPU optimization, and energy-efficient AI.
Reliable and Responsible AI: Multilingual fact verification, hallucination detection and mitigation, trustworthy AI benchmarking, trustworthy AI for scientific discovery.

EDUCATION

University of North Texas (UNT) Aug. 2023 – present
Ph.D. in Computer Science (candidate; expected May 2027)
Advisors: Dr. Yunhe Feng (major), Dr. Kewei Sha (co-major)

University of Birmingham Sept. 2021 – Dec. 2022
*M.Sc. in Data Science, awarded with **Distinction***

Xiamen University Sept. 2017 – Aug. 2021
*B.Sc. in Computer Science and Technology, with **Honours***

RESEARCH GRANTS AND SUPPORTS

Research Grant – Amazon Research Awards (ARA) 2026
Role: Proposal co-writer, PI: Dr. Yunhe Feng
"TileRoute-Trainium: Tile-Level Mixed-Precision Kernels for Long-Context LLM Inference"

Computing Resource – Argonne Leadership Computing Facility (ALCF) 2024 – present
Role: PI, 5,000-node-hour per allocation
"Large Language Models' Hallucination Detection and Mitigation Ability across Multi Languages"

Funded Training – Argonne National Laboratory 2024
Role: Selected Participant, Est. value: ∼$2,500
"Argonne Training Program on Extreme-Scale Computing (ATPESC) 2024"

PUBLICATIONS

Seven peer-reviewed publications (four as first or co-first author) and five manuscripts under review (four as first author). **Bold** marks my name; * denotes equal contribution.

Peer-Reviewed Publications

[1] **Hanzhi Zhang**, Sumera Anjum, Heng Fan, Weijian Zheng, Yan Huang, and Yunhe Feng. Poly-FEVER: A Multilingual Hallucination Detection Benchmark. *Findings of the Association for Computational Linguistics (EMNLP 2026)*. [Paper].

[2] Bing Fan*, Minghao Li*, **Hanzhi Zhang***, Shaohua Dong, Naga Prudhvi Mareedu, Weishi Shi, Yunhe Feng, Yan Huang, and Heng Fan. Towards Visual Query Segmentation in the Wild. *Advances in Neural Information Processing Systems (NeurIPS 2026)*. [Paper].

[3] Mingchen Li, **Hanzhi Zhang**, Heng Fan, Junhua Ding, and Yunhe Feng. Harmful Factuality: LLMs Correcting What They Shouldn't. *Findings of the Association for Computational Linguistics (EACL 2026)*. [Paper].

[4] **Hanzhi Zhang**, Heng Fan, Kewei Sha, Yan Huang, and Yunhe Feng. DAM: Dynamic Attention Mask for Long-Context Large Language Model Inference Acceleration. *Findings of the Association for Computational Linguistics (ACL 2025)*. [Paper].

[5] **Hanzhi Zhang**, Harold Lucero, Kewei Sha, Heng Fan, Song Fu, and Yunhe Feng. D2SO: Detecting Distant and Small Objects for Vision-Based Vehicle Autonomous Systems. *Proceedings of the IEEE International Conference on Mobility, Operations, Services and Technologies (IEEE MOST 2025)*. [Paper].

[6] Sumera Anjum, **Hanzhi Zhang**, Wenjun Zhou, Eun Jin Paek, Xiaopeng Zhao, and Yunhe Feng. HALO: Hallucination Analysis and Learning Optimization to Empower LLMs with Retrieval-Augmented Context for Guided Clinical Decision Making. *Proceedings of the IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE 2025)*. [Paper].

[7] Tong Xie, **Hanzhi Zhang**, Shaozhou Wang, Yuwei Wan, Imran Razzak, Chunyu Kit, Wenjie Zhang, and Bram Hoex. ByteScience: Bridging Unstructured Scientific Literature and Structured Data with Auto Fine-Tuned Large Language Model in Token Granularity. *Proceedings of the IEEE International Conference on Data Mining Workshops (ICDMW 2024)*. [Paper].

Manuscripts Under Review

[8] **Hanzhi Zhang**, Qiao Zhang, Qinglei Cao, Heng Fan, Yan Huang, Kewei Sha, and Yunhe Feng. AlignQuant: Tile-Aligned Mixed-Precision Quantization for Efficient LLM Generation. Under review at the International Conference on Learning Representations (ICLR 2027). [Paper].

[9] **Hanzhi Zhang**, Qiao Zhang, Qinglei Cao, Heng Fan, Yan Huang, Kewei Sha, and Yunhe Feng. TileMix: Tile-Centric Mixed-Precision Attention for LLM Inference Acceleration. Under review at The ACM Web Conference (WWW 2027). [Paper].

[10] **Hanzhi Zhang**, Qiao Zhang, Qinglei Cao, Heng Fan, Yan Huang, Kewei Sha, and Yunhe Feng. TileQuant: Tile as the Precision-Allocation Unit for LLM Mixed-Precision Post-Training Quantization. Under review at the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2027).

[11] **Hanzhi Zhang**, Qiao Zhang, Qinglei Cao, Heng Fan, Yan Huang, Kewei Sha, and Yunhe Feng. TriAttention: Unified Sparse Mixed-Precision Attention for Long-Context LLM Inference. Under review at the Conference on Machine Learning and Systems (MLSys 2027).

[12] Qiao Zhang, Rabab Alomairy, **Hanzhi Zhang**, Yunhe Feng, George Bosilca, and Qinglei Cao. From FP64 to 1-Bit: Scalable and Efficient Multi-Precision Matrix Multiplication. Under review at the ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2027).

TEACHING AND MENTORING EXPERIENCES

Teaching Assistant, CSCE 4201 Introduction to Artificial Intelligence Fall 2026
Undergraduate course, 74 students

- Independently designed and delivered three 3-hour lectures on CNNs, NLP, and Transformers.
- **92.6%** of survey respondents expressed interest in attending another lecture by me.

Teaching Assistant, CSCE 5300 Introduction to Big Data and Data Science Fall 2024, Spring 2025
Graduate core course, 79 students (Fall 2024), 64 students (Spring 2025)

- Received an overall teaching evaluation rating of **"Excellent"** across both semesters.
- Managed grading and office hours; coordinated randomized-question exams and analyzed score distributions.

Guest Lecturer, NSF REU Program Summer 2026

- Gave a lecture "Long-Context Evaluation of Large Language Models".

Academic Coordinator and Research Mentor, NSF REU Program Summer 2026

- Coordinated research training for **9 undergraduates** (6 from MSIs, 1 from an HBCU, and 1 from an R2 university), including weekly research talks and ongoing mentoring.

Student Research Mentoring and Academic Guidance 2023 – present
Responsible AI Lab, UNT

- Provided research and academic guidance to **8 graduate students, 2 undergraduates, and 1 Texas Academy of Mathematics and Science (TAMS) student**.
- Mentored two students on research projects resulting in publications: **Poly-FEVER** (EMNLP 2026 (Findings)) and **D2SO** (IEEE MOST 2025).
- Developed HPC and Docker tutorials enabling junior researchers to conduct experiments independently.

HONORS AND AWARDS

Top Reviewer, NeurIPS 2026 — Oct. 2026

Student Travel Grant Award, IEEE MOST 2025 — Apr. 2025

First Place, Poster presentation, UNT Engineering Showcase — Mar. 2024

PROFESSIONAL SERVICES

Conference Reviewer

- **Machine learning and NLP:** WWW 2027; ICLR 2027; NeurIPS 2026 (*Top Reviewer*); ICML 2026; ACL Rolling Review 2026 (May, August, and October cycles)
- **Data mining and systems:** CIKM 2025; ICDM 2024; IEEE ICPADS 2023

Journal Reviewer

- IEEE IoT-J 2026; IEEE JBHI 2026; Smart Health 2026; Multimedia Tools and Applications 2026

External Reviewer

- ACM SEC 2026; ICDM 2025; WWW 2024; AAAI 2024; WACV 2024; ICNC 2024; EISA 2023; Inscrypt 2023

Conference and Workshop Volunteer

Student Volunteer, NSF CISE MSI Aspiring PI Workshop, UNT — Dec. 2024

Student Volunteer, IEEE MOST 2024 — May 2024

MEMBERSHIPS AND ACTIVITIES

Member, **IEEE Women in Engineering (WIE)** — 2026 – present

Member, **Women in Data Science (WinDS)**, University of North Texas chapter — 2025 – present

Member, **IEEE Young Professionals** — 2025 – present

Graduate Student Member, **Institute of Electrical and Electronics Engineers (IEEE)** — 2025 – present

Student Member, **Association for Computing Machinery (ACM)** — 2024 – present

Participant, **Women-in-Computing session**, IEEE MOST 2025 — Apr. 2025

Participant, **AI Prompt-a-Thon**, UT Arlington and Microsoft — May 2024

INVITED TALKS AND PRESENTATIONS

Invited talk, "Responsible AI," Presidential AI Challenge (Denton ISD and UNT) – audience of educators and industry professionals — May 2026

Invited talk, "Responsible AI: The Hidden Curriculum," AIxHEART – professional development session for middle-school teachers — Oct. 2025

Conference presentation, "DAM: Dynamic Attention Mask for Long-Context LLM Inference Acceleration," ACL 2025 (Findings) — Jul. 2025

Conference presentation, "D2SO: Detecting Distant and Small Objects for Vision-based Vehicle Autonomous Systems," IEEE MOST 2025 — May 2025

Research presentations, LLaVi Lab AI Seminar, UNT (to faculty and Ph.D. students) — 2024 – 2026

Proposal presentation, "Efficient Small Language Models for Edge Devices via Task-Aware Sparse Attention," NSF IUCRC eCAT meeting — Nov. 2024

Poster presentation, "DAM: Dynamic Attention Mask for Long-Context LLM Inference Acceleration," UNT University Research Day — Oct. 2025

Poster presentation, "DAM: Dynamic Attention Mask for Long-Context LLM Inference Acceleration," North Texas Symposium on Generative AI, UT Arlington — Mar. 2025

Poster presentation, "Poly-FEVER: A Multilingual Hallucination Detection Benchmark," Oct. 2024
UNT University Research Day

Poster presentation, "Poly-FEVER: A Multilingual Hallucination Detection Benchmark," Mar. 2024
UNT Engineering Showcase (*First Place*)

Lab research showcase, Responsible AI Lab, UNT CSE Open House 2023 – 2026

PROFESSIONAL EXPERIENCE

Responsible AI Lab, University of North Texas Aug. 2023 – present
Graduate Research Assistant *Denton, TX*

- Lead hardware-aware LLM inference research spanning sparse attention (**DAM**), mixed-precision attention (**TileMix**, **TriAttention**), and quantization (**TileQuant**, **AlignQuant**).
- Led research that established the technical foundation for the lab's **Amazon Research Awards** grant (*TileRoute-Trainium*) and contributed to writing the proposal.
- Developed two research benchmarks: **Poly-FEVER** (77,973 claims, 11 languages) and **VQS-4K** (4,111 videos, >1.3M frames; NeurIPS 2026).
- Conduct large-scale GPU experiments on ALCF **Polaris** and TACC **Lonestar6**.

Oak Ridge National Laboratory (ORNL) 2026 – present
Research Collaborator, Analytics and AI Methods at Scale (AAIMS) Group *Remote*

- Collaborate with Dr. Feiyi Wang, Dr. Emily Herron and Dr. Aditya Kashi on AI-based 3D boundary-to-domain operator learning for partial differential equations (PDEs), targeting **ICML 2027** for submission.

Amazon Research Awards (ARA) 2026 – present
Research Collaborator, TileRoute-Trainium *Project* *Remote*

- Collaborate with Qian Qu (Amazon scientist) on efficient, mixed-precision LLM inference using **AWS Trainium** accelerators.

Argonne National Laboratory – ATPESC 2024 Summer 2024
Selected Participant, Argonne Training Program on Extreme-Scale Computing *Lemont, IL*

- Competitively selected for an intensive two-week program on extreme-scale computing: GPU computing, parallel programming models, and performance optimization on leadership-class supercomputers.